

AI-Based Stroke Phase Detection and Symmetry Analysis in Competitive Swimming Footage: A Proof of Concept

Progress Report

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Abstract—Computer vision analysis in aquatic environments presents unique challenges due to refraction, turbulence, and occlusion, which hinder the accuracy of standard motion tracking systems. This project proposes a non-intrusive AI-based proof of concept system for reliable pixel-to-meter scaling using markers in the background. The core contribution is a novel methodology for dynamic pixel-to-meter scaling using standard lane rope markers as calibration objects, addressing the limitations of fixed-camera setups in varying pool environments. This progress report outlines the development of the modular marker prototypes and the architectural design of the computer vision pipeline. It concludes by defining the evaluation protocols and the remaining milestones required to validate the system’s robustness in real-world scenarios

I. INTRODUCTION

A. Problem Statement

Performance analysis in competitive swimming is traditionally reliant on manual observation or expensive, sensor-based systems that can be intrusive to the athlete. While computer vision offers a non-intrusive alternative, the aquatic environment introduces significant noise. Light refraction at the air-water interface, surface turbulence caused by the swimmer, and occlusions like bubbles and splashes degrade the quality of visual data [1], [9]. Furthermore, camera distances and angles vary significantly between venues, making consistent metric quantification, such as speed in meters/second, difficult without distinct calibration markers.

B. Motivation

There is a distinct lack of standardized, automated systems for extracting swimming metrics from video footage that do not require complex setups. A standard, simple, and well documented system capable of extracting metric data from video with minimal setup could act as a proof of concept, to prove the viability of such systems in aquatic environments. This project focuses on validating feasibility and robustness rather than achieving state-of-the-art performance.

C. Aims and Objectives

The primary aim of this project is to develop a Proof of Concept system that enables the extraction of swimming metrics from video footage. The specific objectives are:

- 1) To design, prototype, and create a modular, under-water marker design that maximizes underwater visibility.
- 2) To create and annotate a small dataset of swimming footage using the designed markers in the background.
- 3) To establish a pipeline for dynamic pixel-to-meter scale extraction using the markers for calibration.
- 4) To calculate simple metrics such as swimmer velocity and stroke length using the calibrated scale.

II. BACKGROUND RESEARCH & LITERATURE REVIEW

Existing literature in sports biomechanics highlights the importance of kinematic metrics like stroke length and propulsion symmetry. Computer vision research has successfully applied deep learning for human pose estimation (HPE) and action recognition in water sports. However, a critical limitation in many CV-based swimming systems is the inability to convert pixel movement into absolute, real-world distances, often relying on relative metrics or requiring external sensors.

This project draws upon two key areas of research:

- **Scale Estimation:** Research on using planar objects or known reference markers for depth and scale estimation is relevant, which we adapt by fixing submerged markers to lane ropes at fixed intervals.
- **Pose Detection & Estimation:** Advances in HPE using CNNs such as OpenPose or HRNet provide a foundation for tracking swimmer joint positions over time, enabling the calculation of stroke metrics once a reliable scale is established.

A. AI in Sports Computer Vision

Computer vision (CV) has fundamentally altered the landscape of sports analytics, transitioning from manual notation to automated, high-fidelity measurement. In the domain of modern sports science, AI-driven systems are now the standard

for extracting kinematic data, estimating pose, and tracking athlete trajectories in real-time.

As noted in [11], AI has emerged as a transformative tool in sports science, enabling objective, high-frequency measurements that surpass the limitations of traditional coaching. In particular, deep learning models have been used to automate event detection, action recognition, and performance prediction across a range of sports such as football, basketball, and athletics. While these advances demonstrate the potential of data-driven methods to tackle complex spatio-temporal patterns, the transfer of such techniques to an underwater setting remains a relatively under-explored domain.

Field sports provide stable visual environments, but aquatic environments present unique noise factors. Consequently, sports analytics has emerged as a critical benchmark for evaluating the robustness and generalization capabilities of AI models. If a system can perform accurately in the chaotic visual domain of a swimming pool, it demonstrates a level of geometric reasoning that far exceeds standard requirements.

B. Swimming as a Stress Test for AI Robustness

While swimming analytics provides value to coaches, from a computer vision perspective, the pool environment serves as an adversarial domain that stress-tests the limits of image processing algorithms. The aquatic setting introduces non-linear distortions that are rarely encountered in terrestrial datasets.

The primary challenge is refraction. Light bending at the air-water interface creates unpredictable geometric distortions, effectively shifting the apparent location of underwater objects relative to the camera. This violates the pinhole camera model assumptions central to many standard calibration techniques. Furthermore, the environment is plagued by dynamic noise. As [9] highlights in their abstract, measurements are “strongly influenced by disturbing elements”, specifically bubbles, splashes, and surface reflections. These elements introduce severe occlusion and high-frequency visual noise that can confuse feature extractors trained on clean data.

Additionally, swimming pools lack standardization. Lane rope designs can vary in colour and sectioned lengths [13], lighting conditions shift rapidly, and turbulence alters structural appearance. This constitutes a significant domain shift problem. For an AI model, a swimming pool is a chaotic, noisy environment, where the only way to achieve successful detection of structural features requires robust techniques.

C. Traditional Computer Vision Approaches

Historically, structural feature detection in swimming relied on handcrafted features and traditional image processing pipelines. These methods operate on the premise that semantic objects such as lane lines can be defined by explicit mathematical properties, for example, sharp intensity gradients or specific colour thresholds.

Common techniques include:

- **Edge Detection & Hough Transform:** Algorithms like Canny or Sobel, combined with voting-based techniques to detect lines.
- **Colour Segmentation:** Early methods relied on handcrafted colour thresholds to isolate the rope’s blue-and-white pattern or the red sections mandated by World Aquatics regulations [13].

However, these methods break down very quickly. The alternating colour bands and non-uniform spacing of blue/white sections vary across pool designs, compromising robustness. As discussed in [4], traditional stereo vision often requires controlled conditions. In a pool, a Canny edge detector cannot distinguish between a lane rope edge and a turbulent wave, and colour segmentation fails when reflections alter the apparent hue.

D. CNN-Based Detection and Segmentation

The limitations of handcrafted features spurred the shift toward Convolutional Neural Networks (CNNs). Unlike traditional CV, CNNs learn hierarchical feature representations, allowing them to encode invariance to local noise like splashes or refraction.

Modern approaches employ CNNs for semantic segmentation of rope pixels, followed by post-processing to merge fragmented segments. Notably, [8] demonstrates that a lightweight U-Net can recover the rope’s position even when the water surface induces significant distortion. Similarly, [12] demonstrated the utility of YOLOv8 for swimmer stroke identification, proving that CNNs can track complex motion even in aquatic footage.

However, while CNNs offer superior robustness, the literature lacks a systematic investigation of how rope design variability impacts detection performance, especially when cameras are positioned at different angles or when the pool is subject to transient occlusions such as parts of the swimmers’ body blocking the lane rope.

E. Pixel-to-Metre Scaling in Computer Vision

Extracting metric data, such as metres per second, from 2D pixel coordinates is a fundamental geometric inference problem. Without a reference scale it is impossible to obtain meaningful real-world data; A velocity of x pixels per second means nothing when the real-world distance described by x pixels changes every frame in a video.

Calibration Challenges

The gold standard requires full camera calibration using a planar reference grid, but this is impractical for broadcast footage. Consequently, systems may rely on alternative scaling methods derived from the environment. [8] bypass the need for explicit intrinsic calibration or planar homography by operating a drone at a consistent height to ensure a uniform scale and perspective. Instead of using lane ropes for pixel-to-meter conversion, the strategy estimates velocity by leveraging the standard 10-metre distance between lane markers as a ground truth reference. While this approach avoids the complexity of

scale calibration, their system assumed a fixed camera height and angle, and relied on specific pre-defined colour thresholds.

Furthermore, World Aquatics' 2025 specifications [13] detail the width and spacing of the red sections of the lane rope which could be critical to acquiring a consistent calibration scale in certain scenarios, such as when the whole length of the lane rope is present in a video's frame. Unfortunately, in situations where this is not the case, different features must be considered.

F. AI Methods for Geometric Reasoning

Recent advancements in AI have moved beyond simple detection to geometric reasoning, that is, inferring 3D structure or metric scale from 2D images.

- Monocular Depth Estimation: Models like MiDaS [2] use deep learning to predict relative depth maps from single images, deriving depth cues that traditional geometry misses.
- Deep Distance Learning: [5] introduced "DisNet," estimating distance directly from bounding box parameters.
- Category-Level Pose Estimation: [7] proposed frameworks that jointly estimate shape and metric scale for objects.

These "geometry-aware" networks suggest a pathway for swimming: instead of relying on brittle manual calibration or planar assumptions [3], a neural network could infer the scale factor by analysing the semantic structure of the markers, treating the scaling factor as a learnable parameter. While geometry-aware networks suggest future extensions, this thesis limits implementation to 2D planar approximations assuming a constant depth for the markers and swimmer.

G. Computer Vision for Swimming Analytics

The end goal of many CV-based swimming studies is to recover motion-specific metrics. [12] Leverages NNs to classify what stroke a swimmer is detected to be swimming. Likewise, [10] proposes a pose-estimation network trained on a large collection of swimming videos to localize key joints in the presence of water reflections. [8] estimates distance travelled by tracking a swimmer's centre of mass, subsequently inferring velocity and average stroke length.

These works illustrate that modern approaches temporal tracking over spatial calibration. The works cited above all agree that surface turbulence and glare can degrade tracking accuracy. However, none of these works address the critical challenge of dynamic pixel-to-metre scaling in detail. Most either assume a fixed camera setup or rely on relative metrics, limiting their applicability across different pool environments.

H. Gap Analysis and Research Need

Reviewing the convergence of CV in sports and swimming analytics reveals distinct gaps, particularly regarding datasets and robust scaling.

Datasets and Benchmarks

The scarcity of publicly available datasets in swimming analytics is a significant barrier. SwimmerNET [9] provides annotated 2-D keypoints but lacks underwater scenes. Moar [12] provides 3-D joint trajectories, relying on markers. Few papers exist that provide swimming analytics, each relying on a different dataset, making cross-comparison difficult.

Open Challenges

The reviewed literature confirms that persistent challenges remain:

- 1) Optical Distortions: Refraction and glare render hand-crafted segmentation fragile [10].
- 2) Variable Environments: Pool designs and lighting vary widely, complicating generalization [8].
- 3) Lack of suitable calibration features: Existing works either assume fixed camera setups or lack robust pixel-to-metre scaling methods [8], [10].

These gaps highlight the need for a robust, generalizable AI pipeline that can detect marker features accurately and extract a stable pixel-to-metre scale. The present thesis builds upon the foundational work of [8] and [10], extending it by developing a multiscale, attention-based segmentation module and integrating a physics-informed homography estimator.

III. PROPOSED SOLUTION: SYSTEM OVERVIEW

The proposed system is partitioned into the following three steps:

A. Marker Calibration

This step is the foundation of the system, establishing a stable axis and scale for all downstream measurements. A CNN-based object detection model will identify the lane rope markers and the swimmer in each frame.

B. Feature Extraction

In this step, the keypoints of the swimmer will be extracted using a pre-trained pose-estimation model fine-tuned on swimming footage. The lane rope markers detected in Step 1 will be used to compute a pixel-to-metre scale for each frame.

C. Metric Calculation

Using the pixel-to-metre scale from Step 1 and the keypoints from Step 2, swimming metrics such as velocity and stroke length will be calculated.

IV. EVALUATION PLAN

The evaluation strategy is two-fold, encompassing a rigorous quantitative assessment of the core technology and a qualitative validation of the final derived metrics.

A. Evaluation Plan

To evaluate the performance of step 1, the detection rate, as well as Intersection over Union (IoU), will be calculated against a manually annotated ground truth dataset. For step 2, the accuracy of the pose estimation will be assessed using standard metrics such as Percentage of Correct Keypoints (PCK) and Mean Per Joint Position Error (MPJPE) against annotated keypoints.

To evaluate step 3, the calculated metrics (velocity, stroke length) will be compared against ground truth measurements obtained from manual timing and distance measurements during controlled swimming trials.

V. CONCLUSIONS INCLUDING EXPECTED OUTCOMES AND DIFFICULTIES/CHALLENGES

This project is expected to deliver a robust proof of concept system demonstrating a novel approach to dynamic pixel-to-metre scaling in swimming footage by proposing a standardized lane rope marker design. The expected outcome is a working pipeline capable of extracting meaningful swimming metrics from video data.

Key challenges and difficulties anticipated include:

- Data Annotation: Obtaining sufficient, frame-by-frame annotated footage to provide ground truth for both marker detection and pose estimation.
- Scale Stability: Ensuring that the pixel-to-metre scale remains stable across frames despite turbulence and occlusions.
- Metric Consistency: Validating that the derived metrics align closely with manual measurements, accounting for potential sources of error in both the video analysis and ground truth data.

In summary, this report outlines the implementation achieved thus far and details the remaining steps for the proposed solution. For the full project schedule, refer to the Gantt chart located in Appendix A.

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VI. APPENDIX A: PROJECT TIMELINE

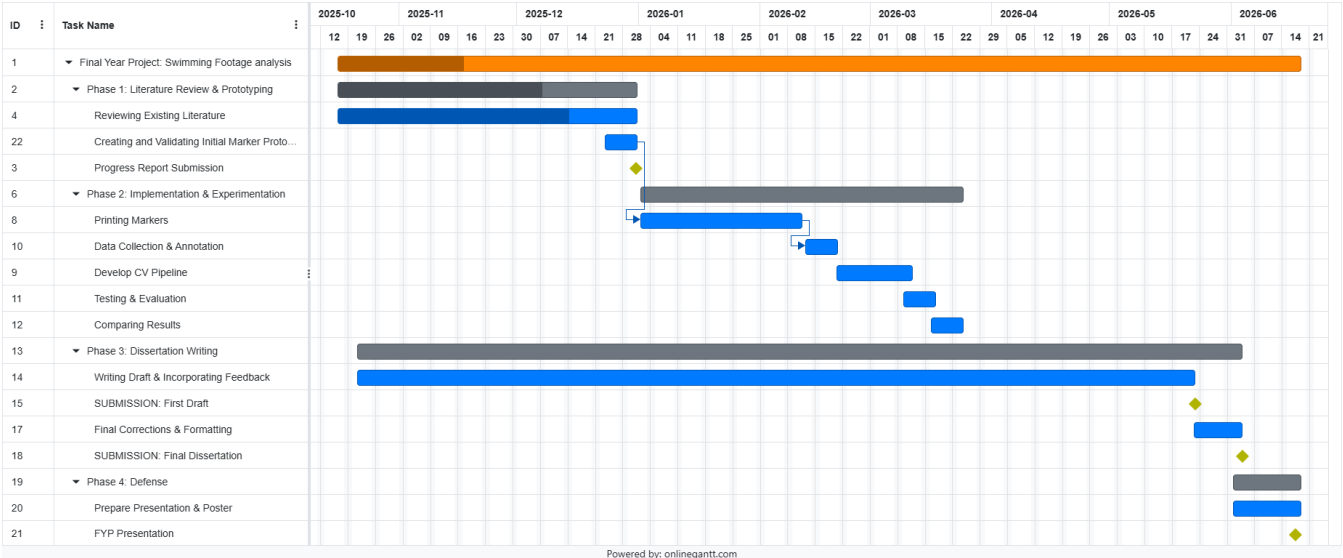


Fig. 1. Project timeline and milestones for the AI-based stroke phase detection system.